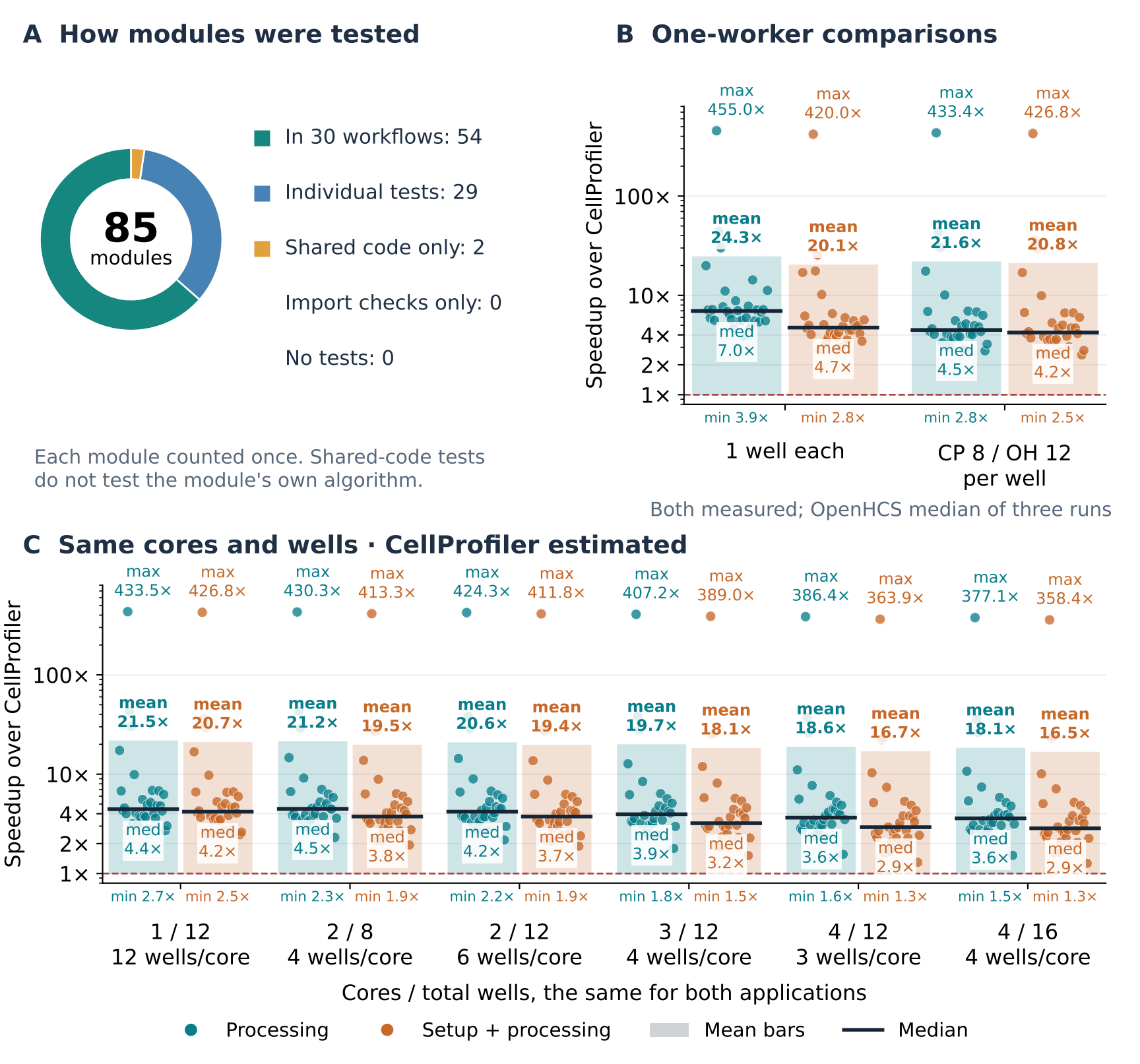
A How modules were tested
85
modules
In 30 workflows: 54
Individual tests: 29
Shared code only: 2
Import checks only: 0
No tests: 0
Each module counted once. Shared-code tests do not test the module's own algorithm.
B One-worker comparisons
Speedup over CellProfiler
max 455.0×
max 420.0×
max 433.4×
max 426.8×
mean 24.3×
mean 20.1×
mean 21.6×
mean 20.8×
med 7.0×
med 4.7×
med 4.5×
med 4.2×
min 3.9×
min 2.8×
min 2.8×
min 2.5×
1 well each
CP 8 / OH 12 per well
Both measured; OpenHCS median of three runs
C Same cores and wells · CellProfiler estimated
Speedup over CellProfiler
max 433.5×
max 426.8×
max 430.3×
max 413.3×
max 424.3×
max 411.8×
max 407.2×
max 389.0×
max 386.4×
max 363.9×
max 377.1×
max 358.4×
mean 21.5×
mean 20.7×
mean 21.2×
mean 19.5×
mean 20.6×
mean 19.4×
mean 19.7×
mean 18.1×
mean 18.6×
mean 16.7×
mean 18.1×
mean 16.5×
med 4.4×
med 4.2×
med 4.5×
med 3.8×
med 4.2×
med 3.7×
med 3.9×
med 3.2×
med 3.6×
med 2.9×
med 3.6×
med 2.9×
min 2.7×
min 2.5×
min 2.3×
min 1.9×
min 2.2×
min 1.9×
min 1.8×
min 1.5×
min 1.6×
min 1.3×
min 1.5×
min 1.3×
1 / 12
12 wells/core
2 / 8
4 wells/core
2 / 12
6 wells/core
3 / 12
4 wells/core
4 / 12
3 wells/core
4 / 16
4 wells/core
Cores / total wells, the same for both applications
Processing
Setup + processing
Mean bars
Median